



Middle East Technical University

Department of Electrical and Electronics Engineering

EE568 – Selected Topics in Electrical Machines

Project 2: Motor Winding Design & Analysis

Author: Elif Topaloğlu

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GitHub Repository:

<https://github.com/odtu/EE568/tree/master/Project2>

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Introduction

1 Q1: Integral-Slot Winding Design

For this question, two winding designs are done namely full-pitched and 11/12 short-pitched windings. The given machine parameters for these windings are:

- Number of poles: $p = 6$
- Number of stator slots: $Q = 72$
- Number of phases: $m = 3$
- Winding configuration: Double-layer

Derived Parameters:

1. Slots per pole per phase (q):

$$q = \frac{Q}{p \cdot m} = \frac{72}{6 \times 3} = 4$$

2. Electrical slot angle (α_e):

$$\alpha_e = \frac{p \times 180^\circ}{Q} = \frac{6 \times 180^\circ}{72} = 15^\circ/\text{slot}$$

3. Mechanical slot angle (α_m):

$$\alpha_m = \frac{360^\circ}{Q} = \frac{360^\circ}{72} = 5^\circ/\text{slot}$$

4. Pole pitch (τ_p):

$$\tau_p = \frac{Q}{p} = \frac{72}{6} = 12 \text{ slots}$$

1.1 Configuration A: Full-Pitched Winding

1.1.1 Winding Diagram

For a full-pitched winding, the coil span equals the pole pitch: $y = \tau_p = 12$ slots, corresponding to 180° electrical. In a double-layer winding, the top layer conductor in slot i is connected to the bottom layer conductor in slot $i + y$. This means the bottom layer of slot i serves as the return side of the coil whose top side resides in slot $i - y$. Since the phase sequence repeats every 12 slots with alternating polarity, the bottom layer ends up identical to the top layer at each slot position.

The phase sequence follows the order $A, -C, B, -A, C, -B$, giving a balanced 3-phase layout across the 24-slot pole-pair:

Table 1: Double-layer full-pitched winding layout (one pole-pair)

Slot	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Top	A	A	A	A	-C	-C	-C	-C	B	B	B	B	-A	-A	-A	-A	C	C	C	C	-B	-B	-B	-B
Bottom	A	A	A	A	-C	-C	-C	-C	B	B	B	B	-A	-A	-A	-A	C	C	C	C	-B	-B	-B	-B

1.1.2 Winding Factor Calculations

The general formulas for the n -th harmonic are:

$$k_{d,n} = \frac{\sin\left(\frac{n \cdot q \cdot \alpha_e}{2}\right)}{q \cdot \sin\left(\frac{n \cdot \alpha_e}{2}\right)} \quad k_{p,n} = \sin\left(\frac{n \cdot y_e}{2}\right) \quad k_{w,n} = k_{d,n} \cdot k_{p,n}$$

where $y_e = 180^\circ$ electrical for full pitch.

Fundamental ($n = 1$):

$$k_{d,1} = \frac{\sin(4 \times 15^\circ / 2 \times 1)}{4 \sin(15^\circ / 2 \times 1)} = \frac{\sin(30^\circ)}{4 \sin(7.5^\circ)} = \frac{0.500}{4 \times 0.131} = \frac{0.500}{0.522} = 0.958$$

$$k_{p,1} = \sin(90^\circ) = 1.000 \quad \Rightarrow \quad k_{w,1} = 0.958$$

3rd Harmonic ($n = 3$):

$$k_{d,3} = \frac{\sin(3 \times 30^\circ)}{4 \sin(3 \times 7.5^\circ)} = \frac{\sin(90^\circ)}{4 \sin(22.5^\circ)} = \frac{1.000}{4 \times 0.383} = \frac{1.000}{1.531} = 0.653$$

$$k_{p,3} = \sin(3 \times 90^\circ) = \sin(270^\circ) = -1.000 \quad \Rightarrow \quad k_{w,3} = -0.653$$

5th Harmonic ($n = 5$):

$$k_{d,5} = \frac{\sin(5 \times 30^\circ)}{4 \sin(5 \times 7.5^\circ)} = \frac{\sin(150^\circ)}{4 \sin(37.5^\circ)} = \frac{0.500}{4 \times 0.609} = \frac{0.500}{2.435} = 0.205$$

$$k_{p,5} = \sin(5 \times 90^\circ) = \sin(450^\circ) = 1.000 \quad \Rightarrow \quad k_{w,5} = 0.205$$

Table 2: Winding factors for the full-pitched configuration.

Harmonic (n)	$k_{d,n}$	$k_{p,n}$	$k_{w,n}$
1 (Fundamental)	0.958	1.000	0.958
3	0.653	-1.000	-0.653
5	0.205	1.000	0.205

1.2 Configuration B: 11/12 Short-Pitched Winding

1.2.1 Winding Diagram

For the 11/12 short-pitched design, the coil span is reduced by one slot: $y = 11$ slots. In electrical degrees:

$$y_e = 11 \times 15^\circ = 165^\circ \text{ electrical}$$

The top layer is identical to the full-pitch case. However, the bottom layer of slot i now carries the return conductor of the coil whose top side is at slot $i - 11$. This shifts the bottom layer by one slot relative to the full-pitch arrangement:

Table 3: Double-layer 11/12 short-pitched winding layout.

Slot	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Top	A	A	A	A	-C	-C	-C	-C	B	B	B	B	-A	-A	-A	-A	C	C	C	C	-B	-B	-B	-B
Bottom	A	A	A	-C	-C	-C	-C	B	B	B	B	-A	-A	-A	-A	C	C	C	C	-B	-B	-B	-B	A

1.2.2 Winding Factor Calculations

The distribution factor does not depend on the pitch which makes the distribution factor identical for both configurations. Only the pitch factor changes:

$$k_{p,n} = \sin\left(\frac{n \times 165^\circ}{2}\right) = \sin(n \times 82.5^\circ)$$

Fundamental ($n = 1$):

$$k_{p,1} = \sin(82.5^\circ) = 0.991 \quad \Rightarrow \quad k_{w,1} = 0.958 \times 0.991 = 0.949$$

3rd Harmonic ($n = 3$):

$$k_{p,3} = \sin(247.5^\circ) = -0.924 \quad \Rightarrow \quad k_{w,3} = 0.653 \times (-0.924) = -0.604$$

5th Harmonic ($n = 5$):

$$k_{p,5} = \sin(412.5^\circ) = \sin(52.5^\circ) = 0.793 \quad \Rightarrow \quad k_{w,5} = 0.205 \times 0.793 = 0.163$$

Table 4: Winding factors for the 11/12 short-pitched configuration.

Harmonic (n)	$k_{d,n}$	$k_{p,n}$	$k_{w,n}$
1 (Fundamental)	0.958	0.991	0.949
3	0.653	-0.924	-0.604
5	0.205	0.793	0.163

1.3 Discussion

The following conclusions can be drawn:

1. Moving from full-pitch to 11/12 short-pitch reduces the fundamental winding factor minimally, from $k_{w,1} = 0.958$ to 0.949 (a decrease of 0.9%). The machine sustains nearly the same torque and back-EMF capability.
2. Third harmonic slightly decreased (7.5%) for short-pitch configuration. However, in a balanced 3-phase machine the triplen harmonics ($n = 3, 9, 15, \dots$) cancel in the *line-to-line* voltage and do not appear in the output regardless of their winding factor. Yet, they may still contribute to stator core losses and noise.
3. Short-pitching reduces $|k_{w,5}|$ from 0.205 to 0.163 (approximately 20% reduction). The pitch factor $k_{p,5}$ drops from 1.000 to 0.793. This is the primary benefit of the 11/12 pitch: it meaningfully attenuates the 5th space harmonic, which is responsible for additional losses in the machine.
4. Short-pitching always involves a trade-off between harmonic suppression and fundamental EMF and torque. In this case the 11/12 pitch is a very efficient compromise, it achieves noticeable 5th harmonic reduction at the cost of less than 1% in fundamental performance.

Table 5: Winding factor comparison between the two configurations.

Harmonic	Full Pitch ($y = 12$)		Short Pitch ($y = 11$)	
	$k_{p,n}$	$k_{w,n}$	$k_{p,n}$	$k_{w,n}$
$n = 1$	1.000	0.958	0.991	0.949
$n = 3$	-1.000	-0.653	-0.92	-0.604
$n = 5$	1.000	0.205	0.793	0.163

2 Q2: Fractional-Slot Winding Design

The assigned design has the following parameters:

- Number of stator slots: $N_s = 15$
- Number of rotor poles: $N_m = 14$
- Number of phases: $m = 3$
- Slots per pole per phase: $N_{spp} \approx 0.357$

2.1 Phase Angles of Induced Voltage per Slot

Using the star-of-slots method, the 360° electrical circle is divided into $2m = 6$ zones of 60 each. Each slot phasor is assigned to the phase zone it falls into; if the phasor falls in a negative zone, the coil side is wound in the reverse direction:

Zone	Range	Assignment
A+	[330, 30)	Phase A, positive
C−	[30, 90)	Phase C, negative
B+	[90, 150)	Phase B, positive
A−	[150, 210)	Phase A, negative
C+	[210, 270)	Phase C, positive
B−	[270, 330)	Phase B, negative

Table 6: Star-of-slots zone assignment (60 zones).

Applying the zone assignment to all 15 slots gives exactly 5 slots per phase, confirming a balanced winding:

Slot	φ_k	Phase	Sign	Slot	φ_k	Phase	Sign
1	0	A	+	9	264	C	+
2	168	A	−	10	72	C	−
3	336	A	+	11	240	C	+
4	144	B	+	12	48	C	−
5	312	B	−	13	216	C	+
6	120	B	+	14	24	A	+
7	288	B	−	15	192	A	−
8	96	B	+				

Table 7: Induced EMF phase angle and phase assignment for each slot (fundamental, $n = 1$).

The resulting slot groupings are:

- **Phase A:** Slots 1(+), 2(−), 3(+), 14(+), 15(−)
- **Phase B:** Slots 4(+), 5(−), 6(+), 7(−), 8(+)
- **Phase C:** Slots 9(+), 10(−), 11(+), 12(−), 13(+)

2.2 Phasor Diagram and Winding Factor

The winding factor for the n -th harmonic is computed as the ratio of the vector sum magnitude to the arithmetic sum magnitude of the slot phasors for one phase:

$$k_{w,n} = \frac{\left| \sum_{\text{Phase}} \vec{e}_{k,n} \right|}{N_s/m}$$

where each phasor $\vec{e}_{k,n}$ has unit magnitude and angle $n \cdot \varphi_k$ (negated for slots wound in the − direction).

Fundamental ($n = 1$): The Phase A phasors and their effective angles are:

Slot	Sign	Raw angle φ_k	Effective phasor angle
1	+	0	0
14	+	24	24
15	−	192	12 (192 − 180)
2	−	168	348 (168 − 180 + 360)
3	+	336	336

Table 8: Phase A phasors for $n = 1$. The five phasors cluster tightly around 0, yielding a high winding factor.

$$k_{w,1} = \frac{|\vec{e}_1 + \vec{e}_{14} + \vec{e}_{15} + \vec{e}_2 + \vec{e}_3|}{5} = \frac{4.783}{5} = \mathbf{0.957}$$

3rd Harmonic ($n = 3$): Each phasor angle is multiplied by 3. The Phase A phasors become 0, 72, 36, 324, 288:

$$k_{w,3} = \frac{3.236}{5} = \mathbf{0.647}$$

5th Harmonic ($n = 5$): Each phasor angle is multiplied by 5. The Phase A phasors become 0, 120, 60, 300, 240:

$$k_{w,5} = \frac{1.000}{5} = \mathbf{0.200}$$

Table 9: Winding factors for harmonics $n = 1, 3, 5$. Values are identical for all three phases.

Harmonic (n)	$k_{w,n}$	Note
1 (fundamental)	0.957	Working harmonic ($p = 7$); high k_w
3	0.647	Zero-sequence; cancels in line-to-line voltage
5	0.200	Balanced, well-attenuated relative to fundamental

2.3 Winding Diagram

The winding is single-layer concentrated: each coil wraps around one tooth. The full slot assignment for all three phases around the stator is:

Table 10: Single-layer concentrated winding slot assignment for all phases.

Slot	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Phase	A	A	A	B	B	B	B	C	C	C	C	C	C	A	A
Direction	+	−	+	+	−	+	−	+	+	−	+	−	+	+	−

Balance verification:

- Each phase occupies exactly 5 slots = $N_s/m = 15/3$. ✓
- The phase resultants at $n = 1$ are at 0, 120, -120 respectively, forming a balanced three-phase set. ✓
- The winding factor is identical for all three phases by symmetry. ✓

2.4 Discussion

1. **High fundamental winding factor:** $k_{w,1} = 0.957$ is comparable to well-optimised distributed windings. This is a key advantage of the 15-slot/14-pole combination and results from the tight angular clustering of the five Phase A phasors (all within ± 24 of the resultant direction).
2. **3rd harmonic — zero-sequence:** For $n = 3$, all three phase resultants are co-phasal (pointing in the same direction). This makes the 3rd harmonic a zero-sequence component: it does not produce a rotating MMF wave and is cancelled in line-to-line voltage and in Δ -connected windings. However, it may still drive circulating currents and iron losses in the stator yoke.
3. **5th harmonic — naturally suppressed:** $k_{w,5} = 0.200$ is small relative to $k_{w,1} = 0.957$ (ratio ≈ 0.21). This is considerably lower than for a typical full-pitch distributed winding, confirming that FSCW inherently attenuates higher-order spatial harmonics.
4. **No sub-harmonics:** Since $\gcd(N_s, N_m) = \gcd(15, 14) = 1$, there are no sub-harmonic MMF components (i.e. no harmonics of order lower than $p = 7$). This eliminates large parasitic radial forces and unbalanced magnetic pull, which is a major advantage of this slot/pole combination.

Conclusion**A Project design parameters****B MATLAB Source Code**

The MATLAB script used for the analytical calculations and plots in Q1 is provided below.